

PROCESS AUDIT REPORT

SECTION 1: AUDIT OVERVIEW & HEADER DETAILS

Organization: Indian Space Research Organisation (ISRO)	Audit Date: 01 / 08 / 2026
Department: Space & Satellite	Audit Standard: ISRO Space Quality Standard
Auditor Name & Designation: Jagriti Singh (Quality Engineer)	Component Name: Titanium Mounting Bracket (Ti-6Al-4V)
Auditee Name & Designation: Senior Quality Engineer, ISRO	Part No. & Batch ID: 3S3-ISRO-TMB

SECTION 2: AUDIT OBJECTIVE & SCOPE

Objective: Our objective is to combine the aspects of daily routine inspection through Kaizen (Japanese philosophy) with Deming Wheel Cycle (PDCA) for proper implementation so that Satellite Titanium Mounting Bracket will achieve zero defect system by eliminating potential system failure before it arises.

Tools Used / Philosophy: PDCA (Deming Wheel Cycle), Kaizen (Japanese Philosophy), Zero Defect System (Philip Crosby), FMEA (System, Process).

SECTION 3: AUDIT CHECKLIST

Checklist ID	Clause / Phase	Requirements	Observations	Status
01	PLAN (Competency & SOP)	- Operator Competency - SOP Technological Standard	- Checked operator training & stamps. - Titanium processing certified. - Verification checked. - Process capability checked [Trend: $C_{pk} + C_p$ checked]	PASS

Chec klist ID	Clause / Phase	Requirements	Observations	Statu s
02	DO (Environment & Equipment)	- Tool Calibration - Environment Management System Standards (ISO 14000, ISO 14001, ISO 14012, ISO 14013) compliance - Material risk & Process validation & deviation	- Dimensional Conformance (Bracket length, thickness, bolt hole found in true specification limit set under SOPs) - MSA performed [Gage R&R Study performed which confirms measurement system acceptable & reproducible]	PASS
03	CHECK (Measurement Capability & Inspection)	- Measurement capability and inspection of two stages (PLAN & DO) - MSA (Measurement System Analysis) - Digital Micrometer & Vernier Caliper Inspection - Environmental degradation detected: Zero surface oxidation inspection	Inspection performed for both stages for their requirements & observations using Six Sigma, System, FMEA.	PASS
04	ACT (CAPA & 5S Integration)	Sort (Seiri) & Set in order (Seiton): Non-inspected and inspected bracket has been separated by red & green visual bins. Shine & Standardize: Inspection table layout re-designed with standard color code. Sustain: Daily 5S audit	5S Methodology properly implemented. - Placement of un-inspected & fully inspected Titanium brackets properly identified.	PASS

SECTION 4: NC LOG & CAPA TRACKING

NC Reference ID: NC-TB-107

NC Severity Level: Minor NC

Description: Risk of mix-up of un-inspected & inspected Titanium brackets found during check & inspection stage.

Root Cause Analysis (RCA): Due to unavailability & tagging of visual tags as per SOP standards.

CAPA: Automated Poka-Yoke / AI sensors are used for visual tagging.

Target Completion Date: 01 / 08 / 2026

Action Owner / Responsibility: Jagriti Singh

SECTION 5: AUDITOR CONCLUSION & FINAL DISPOSITION

Overall Audit Assessment Summary:

- Properly identified failure risk with System & Process FMEA.
- Six Sigma & SQC are applied for reducing variation of specification.
- PDCA cycle and 5S methodology are applied for Audit Checklist & Findings stage.
- Titanium Bracket manufacturing process satisfies ISRO Space Quality Standard.
Integrated Quality Management tools & philosophy to ensure high repeatability & zero defect delivery.

Final Audit Status: [✓] **APPROVED**

Auditor Signature: Jagriti Singh

Date: 01 / 08 / 2026